CSC 1102: Structured and Object-Oriented Programming

School Store Inventory Management System

Group details

Group name: _____

Group members: _____

Group leader: _____

GitHub repository: _____

1. Application Description

The School Store Inventory Management System is a beginner-level Java desktop application for managing stock items in a school store. It uses Java Swing for the graphical user interface and stores item objects in an array while the program is running.

2. Main Features

• Add items with a name, category, quantity and unit price.

• View stock items in a table and search by item name.

• Restock items and issue available stock.

• Prevent invalid quantities and prevent issuing more stock than is available.

• Calculate the total value of all stock.

3. Classes and Their Purpose

Class	Purpose
Item	Represents one item. Holds private attributes, a constructor, getters/setters, restock and issue methods, and a valu
StoreInventory	Manages an array of Item objects. Adds and searches items and calculates the total inventory value.
InventoryManagementGUI	Builds the Swing window, reads user input, handles button actions, and displays inventory information.

4. How the Classes Interact

InventoryManagementGUI creates a StoreInventory object. When the user adds an item, StoreInventory creates an Item object and stores it in its Item array. The GUI calls methods on StoreInventory to search and display items. For restocking and issuing, the GUI finds the selected Item and calls its restock() or issueStock() method.

5. Java Concepts Demonstrated

The project demonstrates classes and objects, private fields, constructors, getters and setters, method calls, arrays, if/else, switch, for and while loops, input validation, and a Swing GUI with event listeners. The project does not require a database.

6. Running the Application

With a JDK installed, open a terminal in the project folder and run:

```
javac -d out src/*.java
java -cp out InventoryManagementGUI
```

Before submission, fill in the group details and GitHub repository, add screenshots of the running application if available, and verify the final files against the lecturer's submission instructions.